A QUARTIC OVERLAP CERTIFICATE AND EFFECTIVE FINITE CYLINDER COVERS OF A REGULAR TETRAHEDRON

Abstract. We give an explicit finite version of the angular cylinder construction for a regular tetrahedron, using the secant axes and matching angular boundaries of the OpenAI family-100 manuscripts. A polynomial simultaneous-failure identity permits a quartic radial correction $9e^4/16$. An explicit perturbation bound transfers this certificate to every finite positive-width angular partition and yields compact, nondegenerate triangular perpendicular bases covering the entire closed tetrahedron. For a uniform partition into N intervals, the normalized area differs from its limiting integral by at most $2e/N$ for $0 < e \le 1/10$. In particular, $N \ge \lceil 480/e \rceil$ guarantees a saving of at least $e^2/240$ below one half. On the schedule $N = \lceil 2/e^2 \rceil$ the actual area is $1/2 - e^2/60 + e^3/2 + O(e^4)$, with an explicit quartic error bound. An exact rational certificate places a 20,000-cylinder example in [0.499921585997001674, 0.499921585997001774] times the minimum projection area. The comparison is with identified published recipes on their explicit sufficient ranges and prescribed count conventions; no global optimality or domination of every possible generic-method instance is asserted.

1. Scope, geometry, and the main theorem

The existence of a finite triangular-cylinder cover below half the minimum projection area is already established in the OpenAI family-100 manuscripts [1, 2, 3, 4]. This note uses the angular paper's secant directions, shared sector boundaries, and first-crossing selection [1]. Its contribution is the quartic overlap identity in Lemma 3.2 and the explicit finite perturbation estimate that follows it. These change the proved cap margins and parameter ranges within an existing geometric construction. In particular, neither the secant sectors themselves nor the original counterexample to the half-area conjecture are claimed as new here.

All four source manuscripts and the cited formal source files are pinned to the OpenAI repository revision `1581d4cbdd69a0faf34b2089614fe2a814316a6a`. The public source manifest records all 91 snapshot hashes. The complete package, including supplementary proofs and exact arithmetic code, is available at `shjvandekamp/math`, topic `cylinder-covering`.

Definition 1.1. A cylinder with axis $v \neq 0$ and compact perpendicular base $B \subset v^\perp$ is the closed set $B + \mathbb{R}v$. Its cost is the ordinary two-dimensional area of B. The cost of a finite list of cylinders is the sum of its base areas, with multiplicity. For a convex body K, let $A_{\min}(K)$ be its minimum orthogonal projection area over all directions.

Put $H = \sqrt{2}$ and fix the edge-two regular tetrahedron

$$K = \{(x, y, Hs) : 0 \le s \le 1,\ |x| \le 1 - s,\ |y| \le s\}. \tag{1}$$

Its vertices are $(\pm 1, 0, 0)$ and $(0, \pm 1, H)$, so every edge has length two. The two horizontal edge directions are the x- and y-axes. Angles below are measured from the corresponding unoriented horizontal edge line.

Lemma 1.2 (Projection normalization)**.** *The tetrahedron in* (1) *satisfies* $A_{\min}(K) = H$.

Proof. The four outward face area vectors are $(0, \pm H, -1)$ and $(\pm H, 0, 1)$. For a unit vector v, the elementary polytope projection formula therefore gives

$$A(v) = \max(H|v_x|, |v_z|) + \max(H|v_y|, |v_z|).$$

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Indeed, half the sum of the absolute scalar products of the face area vectors with v counts the projected area: almost every line parallel to v through the projection meets two faces, and projected edges have area zero. Set $U = \max(|v_x|, |v_z|/H)$ and $V = \max(|v_y|, |v_z|/H)$. Then

$$(U+V)^2 \geq v_x^2 + v_y^2 + 2v_z^2/H^2 = 1.$$

Thus $A(v) = H(U+V) \geq H$, with equality for the x-axis. □

Throughout the note an area ratio means total cylinder cost divided by H. Define the polynomials

$$\begin{aligned} &G(q) = \frac{1+q^2}{2}, \qquad D(q) = \frac{1-q^2}{2}, \qquad h(q) = q^2 G(q), \\ &D_0(q) = G(q)^2 + 2q^2, \qquad C_e(q) = 1 + e^2 h(q) + \frac{9}{16}e^4, \qquad M_* = 1 + e^2 + \frac{9}{16}e^4. \end{aligned} \tag{2}$$

In particular, $1 \leq C_e(q) \leq M_*$ on $[-1, 1]$.

Theorem 1.3 (Effective triangular covers)**.** *For every integer $N \geq 1$ and $0 < e < 1$, the explicit construction in Section 2 gives exactly $2N$ labeled cylinders with compact, nondegenerate triangular perpendicular bases covering every point of the closed tetrahedron K. On the equal grid, their maximum axis-angle tangent is*

$$\sqrt{3}\, e(1 - 1/N). \tag{3}$$

Write $R_{\mathrm{sec}}(e, N)$ for their normalized cost and put

$$J(e) = \frac{1}{4}\int_{-1}^{1} \frac{C_e(q)^2}{\sqrt{1 + e^2 D_0(q)}}\, dq. \tag{4}$$

For $0 < e \leq 1/10$ the following hold:

$$|R_{\mathrm{sec}}(e, N) - J(e)| \leq \frac{2e}{N}, \tag{5}$$

$$N \geq \lceil 480/e \rceil \quad \Longrightarrow \quad R_{\mathrm{sec}}(e, N) \leq \frac{1}{2} - \frac{e^2}{240}, \tag{6}$$

$$N \geq \left\lceil 2/e^2 \right\rceil \quad \Longrightarrow \quad R_{\mathrm{sec}}(e, N) \leq \frac{1}{2} - \frac{e^2}{60} + \frac{e^3}{2} + \frac{9}{2}e^4. \tag{7}$$

Consequently, the count $2N = 2\lceil 2/e^2 \rceil$ gives

$$R_{\mathrm{sec}}(e, N) \leq \frac{1}{2} - \frac{73}{15000}e^2 < \frac{1}{2} \qquad (0 < e \leq 1/50). \tag{8}$$

On this specified count schedule the actual asymptotic cost is

$$R_{\mathrm{sec}}(e, \left\lceil 2/e^2 \right\rceil) = \frac{1}{2} - \frac{e^2}{60} + \frac{e^3}{2} + O(e^4). \tag{9}$$

Finally, at $e = 1/10$ and $N = 10000$, exact rational bounds give

$$\begin{aligned} 0.499921585997001674 &\leq R_{\mathrm{sec}}(1/10, 10000) \\ &\leq 0.499921585997001774 < 0.499922. \end{aligned} \tag{10}$$

The coverage assertion is proved for arbitrary positive-width partitions, with local padding. The quantitative estimates above concern the equal grid; a nonuniform version of (5) is given below. The coefficient $1/2$ in (9) describes this recipe, not an optimum over all cylinder covers or all compatible sector constructions.

2. The explicit finite construction

Let $\mathcal{P}$ be a partition of $[-1,1]$ into N closed intervals $I_i = [l_i, r_i]$ of positive width, in their natural order, with disjoint interiors. Set

$$d_i = r_i - l_i, \quad a_i = d_i/2, \quad c_i = (l_i + r_i)/2, \quad A_i = \frac{1 + l_i r_i}{2} = G(c_i) - \frac{a_i^2}{2}. \tag{11}$$

For $0 < e < 1$, define

$$K_0 = M_*(e + e^2/2) + 2e^2, \qquad \delta_i = \frac{K_0 a_i + (M_* e^2/2) a_i^2}{1 - e}, \qquad M_i = \max_{q \in I_i} C_e(q) + \delta_i. \tag{12}$$

The maximum is explicitly attained at the endpoint of largest absolute value, since h is even and increasing as a function of $|q|$.

Take the intercept triangles

$$\begin{aligned} P_i &= \{(0, qt, Ht) : q \in I_i,\ 0 \le t \le M_i/2\}, \\ Q_i &= \{(pt, 0, H(1-t)) : p \in I_i,\ 0 \le t \le M_i/2\}, \end{aligned} \tag{13}$$

and respectively the axes

$$v_i = (1, eA_i, Hec_i), \qquad w_i = (-eA_i, 1, Hec_i). \tag{14}$$

The actual perpendicular bases are the orthogonal projections of P_i onto $v_i^\perp$ and of Q_i onto $w_i^\perp$. The first coordinate of v_i and the second coordinate of w_i are one. Projection is therefore injective on the corresponding intercept plane, and neither positive-area triangle degenerates. Projection also changes every point by a multiple of the axis, so it leaves the associated cylinder unchanged. This defines the promised list of $2N$ compact triangular cylinders.

For the equal grid $d = 2/N$, every δ_i equals

$$\delta(e, d) = \frac{d\{M_*(e + e^2/2) + 2e^2 + M_* e^2 d/4\}}{2(1 - e)}. \tag{15}$$

In the angular paper's physical notation, $\tau = 2e$, $\alpha_i = A_i/2$, and $\beta_i = c_i/2$. Thus (14) are exactly its secant axes, rather than new directions.

Proposition 2.1 (Exact cost and finite angle)**.** *The normalized cost of the construction is*

$$R_{\mathrm{sec}}(e; \mathcal{P}) = \frac{1}{4} \sum_i \frac{d_i M_i^2}{\sqrt{1 + e^2(A_i^2 + 2c_i^2)}}. \tag{16}$$

For the equal grid its maximum axis-angle tangent is (3)*, including the case* $N = 1$*.*

Proof. Each intercept triangle has area $H d_i M_i^2/8$. Orthogonal projection multiplies this by the absolute scalar product of the intercept-plane unit normal with the unit axis, namely $(1 + e^2(A_i^2 + 2c_i^2))^{-1/2}$. Summing the two families and dividing by H proves (16).

The squared tangent from the corresponding horizontal edge is $e^2(A_i^2 + 2c_i^2)$. On an equal grid this increases with $|c_i|$: writing $z = c_i^2$, its factor is $((1 + z)/2 - a_i^2/2)^2 + 2z$, whose derivative is $A_i + 2 > 0$. At either outer interval, $A_i = |c_i| = 1 - 1/N$. This proves the formula. For $N = 1$, both A_i and c_i vanish, so the directions are the horizontal equality directions themselves. □

3. Closed-body selection and the overlap certificate

We first establish angular selection at every point, without assuming a monotone root map and without dividing by a possibly zero radius. This is the shared-boundary first-crossing argument of [1], included here with its endpoint cases.

Lemma 3.1 (First crossing on a finite partition)**.** *For every $(x, y, Hs) \in K$ there are indices i, j and nonnegative numbers t_0, s_0 such that*

$$\begin{aligned} t_0 &= s - exc_i, & y_0 &= y - exA_i, & l_i t_0 \le y_0 \le r_i t_0,\\ s_0 &= 1 - s + eyc_j, & x_0 &= x + eyA_j, & l_j s_0 \le x_0 \le r_j s_0. \end{aligned}$$

If either radius is zero, a cylinder of the construction already covers the point. Otherwise, putting

$$\rho = 2t_0, \quad \sigma = 2s_0, \quad q = y_0/t_0 \in I_i, \quad p = x_0/s_0 \in I_j, \quad X = 2x, \quad Y = 2y,$$

gives

$$Y = \rho q + eXA_i, \qquad X = \sigma p - eYA_j, \qquad \rho + \sigma = 2 + e(Yc_j - Xc_i). \tag{17}$$

Proof. Write $q_0 = -1 < q_1 < \cdots < q_N = 1$ for the partition nodes. For $q \in I_i$, let $L_i(q) = A_i - qc_i$. Direct substitution at either endpoint gives $L_i(q) = D(q)$. The sector boundaries consequently match at common nodes. Moreover, L_i interpolates the two nonnegative endpoint values of D, hence $L_i(q) \ge 0$ throughout its interval.

At the nodes put $F_k = sq_k + exD(q_k)$. Since $F_0 = -s \le y \le s = F_N$, there is a first $k \ge 1$ for which $F_k \ge y$. If $k = 1$, $F_0 \le y$ already; if $k > 1$, minimality gives $F_{k-1} < y$. Thus in both cases $F_{k-1} \le y \le F_k$. Choose $I_i = [q_{k-1}, q_k]$. Replacing $D(q)$ at these endpoints by $A_i - qc_i$ gives exactly

$$l_i(s - exc_i) \le y - exA_i \le r_i(s - exc_i).$$

The outer inequality and $r_i - l_i > 0$ imply $t_0 = s - exc_i \ge 0$. This also handles $y = F_0$; no monotonicity of the remaining node values has been used.

For the upper family repeat the argument with $E_k = (1 - s)q_k - eyD(q_k)$, whose endpoint values bracket x because $|x| \le 1 - s$. It produces the second row of inequalities and $s_0 = 1 - s + eyc_j \ge 0$.

If $t_0 = 0$, the bracketing forces $y_0 = 0$, and the original point is xv_i, on the cylinder through the tip $(0, 0, 0)$ of P_i. If $s_0 = 0$, then $x_0 = 0$, and the point is $(0, 0, H) + yw_j$, on the cylinder through the tip of Q_j. In the remaining case both divisions in the statement are legitimate, and rearranging the definitions gives (17). Every step uses closed inequalities, so this proof covers angular nodes, all faces and edges, and all four vertices. □

For the positive-radius case of Lemma 3.1, put

$$\begin{aligned} \Lambda &= 1 + e^2 A_i A_j,\\ w_\rho &= 1 - eqc_j + e^2 A_j(A_i - qc_i),\\ w_\sigma &= 1 + epc_i + e^2 A_i(A_j - pc_j). \end{aligned} \tag{18}$$

The cell data satisfy $0 \le A_i, A_j \le 1$, and the interpolation just proved gives $L_i, L_j \ge 0$. Therefore

$$w_\rho, w_\sigma \ge 1 - e > 0. \tag{19}$$

Solving the first two equations in (17) yields

$$X = \frac{\sigma p - e\rho q A_j}{\Lambda}, \qquad Y = \frac{\rho q + e\sigma p A_i}{\Lambda}.$$

Substituting these into its third equation gives the exact elimination identity

$$w_\rho \rho + w_\sigma \sigma = 2\Lambda. \tag{20}$$

Lemma 3.2 (Quartic certificate in physical coordinates)**.** *For $p, q \in [-1, 1]$, put*

$$N_0 = 1 + e^2 G(p)G(q), \quad A_- = 1 - epq + e^2 G(p)D(q), \quad B_+ = 1 + epq + e^2 G(q)D(p).$$

Then

$$\begin{aligned} A_-C_e(q)+B_+C_e(p)-2N_0 = \frac{e^2}{2}\left[p^2-q^2+\frac{epq(1+p^2+q^2)}{2}\right]^2 \\ +e^4G(p)G(q)\{p^2D(p)+q^2D(q)\} \\ +\frac{e^4}{8}\{9-p^2q^2(1+p^2+q^2)^2\} \\ +\frac{9}{16}e^6\{G(p)D(q)+G(q)D(p)\}\geq 0. \end{aligned} \tag{21}$$

Proof. Write $\Delta = p^2-q^2$ and $Z = 1+p^2+q^2$. Expanding the left side before completing the square gives

$$\frac{e^2\Delta^2}{2}+e^3pq\{h(p)-h(q)\}+e^4G(p)G(q)\{p^2D(p)+q^2D(q)\}+\frac{9}{16}e^4(A_-+B_+).$$

Here $h(p)-h(q)=\Delta Z/2$ and $A_-+B_+ = 2+e^2\{G(p)D(q)+G(q)D(p)\}$. Completing the square proves the identity. Every displayed term on the right of (21) is nonnegative. For the only factor requiring a bound, $p^2q^2\leq 1$ and $Z\leq 3$ already suffice; the exact nonnegative decomposition is

$$9-p^2q^2Z^2 = 9\{(1-p^2)+p^2(1-q^2)\}+p^2q^2\{(1-p^2)+(1-q^2)\}(3+Z).$$

The plus sign inside the square is the sign for the physical coordinates (1) and (17). □

Lemma 3.3 (Finite secant perturbation)**.** *For the selected intervals and parameters above,*

$$w_\rho C_e(q)+w_\sigma C_e(p)-2\Lambda \geq -K_0(a_i+a_j)-\frac{M_*e^2}{2}(a_i^2+a_j^2). \tag{22}$$

Consequently,

$$w_\rho\{C_e(q)+\delta_i\}+w_\sigma\{C_e(p)+\delta_j\}\geq 2\Lambda. \tag{23}$$

Proof. Writing $b = q-c_i$, define $E_i(q) = (a_i^2-b^2)/2$. Elementary secant identities give

$$L_i(q) = D(q)-E_i(q), \qquad 0\leq E_i(q)\leq a_i^2/2, \qquad |A_i-G(q)|\leq a_i. \tag{24}$$

For the last bound,

$$|A_i-G(q)| = \left|-c_ib-\frac{a_i^2+b^2}{2}\right| \leq |c_i|a_i+a_i^2\leq a_i,$$

using $|c_i|+a_i\leq 1$. The same statements hold for $p\in I_j$. Since $D\leq 1/2$ and $G, A_i, A_j\leq 1$, they imply

$$|w_\rho - A_-|\leq ea_j+\frac{e^2}{2}(a_j+a_i^2), \tag{25}$$

$$|w_\sigma - B_+|\leq ea_i+\frac{e^2}{2}(a_i+a_j^2), \tag{26}$$

$$|\Lambda-N_0|\leq e^2(a_i+a_j). \tag{27}$$

For example, the quadratic term in the first estimate is bounded by expanding

$$A_jL_i(q)-G(p)D(q) = [A_j-G(p)]D(q)-A_jE_i(q).$$

The linear term is bounded by $e|q|\,|c_j-p|\leq ea_j$. The second estimate is identical with the roles exchanged, and the last follows from $A_iA_j-G(q)G(p) = (A_i-G(q))A_j+G(q)(A_j-G(p))$.

Multiply (25) and (26) by the respective caps, use $C_e\leq M_*$, and account for twice the denominator error. Lemma 3.2 then gives (22), with exactly the K_0 in (12). Finally,

$$w_\rho\delta_i+w_\sigma\delta_j\geq (1-e)(\delta_i+\delta_j) = K_0(a_i+a_j)+\frac{M_*e^2}{2}(a_i^2+a_j^2).$$

Adding this to (22) proves (23). □

Proof of the coverage assertion in Theorem 1.3. Choose the two selected intervals using Lemma 3.1. Its zero-radius cases already give coverage. Otherwise, the point belongs to the lower selected cylinder whenever $\rho \le M_i$, and to the upper selected cylinder whenever $\sigma \le M_j$, by (13). If both cylinders missed it, then $\rho > M_i \ge C_e(q) + \delta_i$ and $\sigma > M_j \ge C_e(p) + \delta_j$. The strictly positive weights in (19) would give

$$w_\rho \rho + w_\sigma \sigma > w_\rho\{C_e(q) + \delta_i\} + w_\sigma\{C_e(p) + \delta_j\} \ge 2\Lambda,$$

contradicting (20). Hence one of the two closed cylinders covers the point. Equality at a cap boundary is included. The argument applies to every point of K and every positive-width partition, without an exceptional boundary set. □

Remark 3.4 (Compatible caps beyond constant triangles). The same proof applies to a cellwise cap $R_i(q)$ whenever $R_i(q) \ge C_e(q) + \delta_i$. Replace $0 \le t \le M_i/2$ in (13) by $0 \le t \le R_i(q)/2$. If the resulting intercept sets are compact, their projections are compact cylinder bases; they need not be triangles. Constant maxima were used above to obtain triangular bases and the simple exact area formula.

4. The limiting integral with an explicit remainder

The next elementary calculation supplies every constant needed for the finite area estimates. It is included so that no asymptotic assertion is used as an effective error bound.

Lemma 4.1. *For $0 < e \le 1/10$,*

$$J(e) \le \frac{1}{2} - \frac{e^2}{60} + \frac{929}{1260} e^4 + 4e^6. \tag{28}$$

Moreover, $J(e) = 1/2 - e^2/60 + (929/1260)e^4 + O(e^6)$ as $e \downarrow 0$.

Proof. Let $u = e^2$ and $b = 9/16$. Taylor's theorem, using the negative third derivative, gives for every $z \ge 0$

$$(1+z)^{-1/2} \le 1 - z/2 + 3z^2/8.$$

Multiply this by

$$C_e(q)^2 = 1 + 2h(q)u + (h(q)^2 + 2b)u^2 + 2bh(q)u^3 + b^2u^4.$$

The constant, linear, and quadratic coefficients of the product integrate exactly to the first three terms of (28). Explicitly,

$$\int_{-1}^{1} h(q)\,dq = \frac{8}{15}, \qquad \int_{-1}^{1} D_0(q)\,dq = \frac{34}{15},$$

so $\frac{1}{4}\int(2h - D_0/2) = -1/60$. Also

$$h^2 - hD_0 + \frac{3}{8}D_0^2 = \frac{3 + 44q^2 + 162q^4 - 52q^6 + 19q^8}{128},$$

whose integral divided by two is $881/2520$. Adding the constant contribution $2b = 9/8$ gives $929/630$ for $2J$, or $929/1260$ for J.

The remaining four coefficients of the upper polynomial are

$$c_3 = \tfrac{3}{4}hD_0^2 - \tfrac{1}{2}D_0(h^2 + 2b) + 2bh,$$
$$c_4 = \tfrac{3}{8}D_0^2(h^2 + 2b) - bD_0h + b^2,$$
$$c_5 = \tfrac{3}{4}bhD_0^2 - \tfrac{1}{2}b^2D_0, \qquad c_6 = \tfrac{3}{8}b^2D_0^2.$$

Using $0 \le h \le 1$, $0 \le D_0 \le 3$, and dropping the negative terms gives respectively the pointwise upper bounds

$$\frac{63}{8}, \quad \frac{1917}{256}, \quad \frac{243}{64}, \quad \frac{2187}{2048}.$$

For $u \le 1/100$,

$$\frac{63}{8} + \frac{1917}{25600} + \frac{243}{640000} + \frac{2187}{2048000000} < 8.$$

The factor $\frac{1}{4}\int_{-1}^{1}$ therefore bounds the full remaining contribution by $4u^3$, proving (28). Taylor's remainder is also uniformly bounded on $q \in [-1,1]$ and $e \le 1/10$, since $0 \le e^2 D_0(q) \le 3/100$. The same coefficient computation then gives the stated two-sided asymptotic expansion. □

5. Finite error, an explicit count, and the cubic term

In this section $0 < e \le 1/10$, the grid is equal, and $d = 2/N$. Write $\widehat{M}_i = \max_{I_i} C_e$, $\delta = \delta(e,d)$, and

$$g(q) = (1 + e^2 D_0(q))^{-1/2}, \qquad g_i^{\sec} = (1 + e^2(A_i^2 + 2c_i^2))^{-1/2}.$$

Lemma 5.1 (All discretization errors)**.** *For every $N \ge 1$,*

$$R_{\sec}(e,N) - J(e) \le \left(M_* + \frac{3}{8}M_*^2\right) e^2 d + M_*\delta + \frac{\delta^2}{2} + \frac{M_*^2 e^2 d^2}{16}, \tag{29}$$

and

$$R_{\sec}(e,N) - J(e) \ge -\frac{3}{8}M_*^2 e^2 d. \tag{30}$$

Proof. There are four changes from the integral to the exact sum. First, C_e has total variation $2e^2$ on $[-1,1]$. Thus

$$\sum_i \int_{I_i} (\widehat{M}_i - C_e(q))\, dq \le 2e^2 d.$$

Since $\widehat{M}_i + C_e(q) \le 2M_*$ and $g \le 1$, replacing the cap by its cell maximum increases the normalized integral by at most $M_* e^2 d$.

Second, $|D_0'(q)| \le 6$ and hence $|g'(q)| \le 3e^2$. Because

$$\sum_i \int_{I_i} |q - c_i|\, dq = d/2,$$

replacing $g(q)$ by $g(c_i)$ changes the integral, in absolute value, by at most $3M_*^2 e^2 d/8$.

Third, with $a_i = d/2$,

$$D_0(c_i) - (A_i^2 + 2c_i^2) = G(c_i)a_i^2 - a_i^4/4 \in [0, a_i^2].$$

The derivative of $(1 + e^2 z)^{-1/2}$ in z has absolute value at most $e^2/2$. Consequently

$$0 \le g_i^{\sec} - g(c_i) \le e^2 d^2/8.$$

Applying this to the unpadded maximum caps gives a nonnegative area increase at most $M_*^2 e^2 d^2/16$.

Finally, adding the padding changes their normalized cost by

$$\frac{1}{4}\sum_i d(2\widehat{M}_i\delta + \delta^2) g_i^{\sec} \le M_*\delta + \delta^2/2.$$

The four estimates prove (29). Every change except the midpoint-weight substitution is nonnegative, which also proves (30). □

Proposition 5.2 (Uniform error and count bound)**.** *Equations (5) and (6) hold. More generally, for an arbitrary positive-width partition with the local padding (12),*

$$|R_{\sec}(e;\mathcal{P}) - J(e)| \le e \max_i d_i. \tag{31}$$

Proof. The elementary bounds

$$M_* < \frac{1011}{1000}, \qquad M_*^2 < \frac{1023}{1000}, \qquad M_* + \frac{3}{8}M_*^2 < \frac{7}{5} \tag{32}$$

hold throughout $e \le 1/10$. We also have

$$\delta \le \frac{3}{4}ed \qquad (0 < d \le 2). \tag{33}$$

For an explicit verification of this last bound, subtract $ed/2$ in (15) and clear its denominator to obtain

$$\delta - \frac{ed}{2} = \frac{ed\{(M_* - 1) + e(M_*/2 + 3 + M_* d/4)\}}{2(1-e)}.$$

Here $(M_* - 1)/e = e + 9e^3/16$ is increasing. Substituting $e = 1/10$, $d = 2$ into the remaining upper coefficient gives

$$\frac{(M_*(1/10) - 1)/(1/10) + M_*(1/10) + 3}{2(1 - 1/10)} < \frac{5}{2}.$$

Thus $0 \le \delta - ed/2 \le (5/2)e^2 d \le ed/4$.

Apply (32), (33), $e \le 1/10$, and $ed \le 1/5$ to (29). Its entire right side is at most

$$\left(\frac{7}{50} + \frac{3033}{4000} + \frac{9}{160} + \frac{1023}{80000}\right) ed = \frac{77383}{80000} ed < ed.$$

The absolute lower error in (30) is smaller than ed as well. This proves (5).

For a nonuniform partition set $d = \max_i d_i$. The same maximum-cap variation estimate is bounded by $2e^2 d$. The midpoint estimate uses $\sum_i d_i^2 \le 2d$, and the secant estimate uses $\sum_i d_i^3 \le 2d^2$. Each local δ_i is at most $\delta(e, d)$, by its explicit positive-coefficient formula. The preceding proof therefore proves (31) too.

Finally,

$$\frac{929}{1260 \cdot 100} + \frac{4}{10000} < \frac{1}{120}.$$

Lemma 4.1 gives $J(e) \le 1/2 - e^2/120$. If $N \ge 480/e$, the finite error $2e/N$ is at most $e^2/240$, which proves (6). □

Proposition 5.3 (The source count schedule)**.** *Equations* (7)*,* (8)*, and* (9) *hold.*

Proof. If $N \ge \lceil 2/e^2 \rceil$ then $d \le e^2$. This improves the padding bound to

$$\delta \le \frac{1}{2}ed + \frac{21}{10}e^2 d \le \frac{71}{100}e^3. \tag{34}$$

To verify it, use $M_* \le 1 + (81/80)e^2$, $d \le e^2$, and $e \le 1/10$ to get

$$M_*(1 + e/2 + ed/4) + 2e \le 1 + \frac{5}{2}e + \frac{11}{10}e^2.$$

For example the higher-order coefficient is bounded by

$$\frac{81}{80} + \left(\frac{1}{4} + \frac{81}{160}\right)\frac{1}{10} + \frac{81}{320 \cdot 1000} < \frac{11}{10}.$$

The exact identity

$$(1-e)(1 + 21e/5) - (1 + 5e/2 + 11e^2/10) = 7e/10 - 53e^2/10 \ge 0$$

now proves (34) after division by $2(1-e)$.

All remaining terms can be bounded by rational constants. In (29) use

$$\left(M_* + \frac{3}{8}M_*^2\right)e^2 d \le \frac{7}{5}e^4, \qquad M_*\delta \le \frac{1}{2}e^3 + \frac{109}{50}e^4,$$
$$\delta^2/2 \le \frac{3}{1000}e^4, \qquad M_*^2 e^2 d^2/16 \le \frac{1}{1000}e^4.$$

For the second inequality, (34) supplies the $e^3/2 + (21/10)e^4$ part, and

$$(M_* - 1)\delta \le \frac{81}{80}e^2\frac{71}{100}e^3 < \frac{2}{25}e^4.$$

The others follow directly from (32), (34), and $e^2 \le 1/100$. With $4e^6 \le e^4/25$, the total quartic coefficient from Lemma 4.1 and these inequalities is

$$\frac{929}{1260} + \frac{7}{5} + \frac{109}{50} + \frac{3}{1000} + \frac{1}{1000} + \frac{1}{25} < \frac{9}{2}.$$

This proves (7). If $e \le 1/50$, then

$$\frac{1}{60} - \frac{e}{2} - \frac{9}{2}e^2 \ge \frac{1}{60} - \frac{1}{100} - \frac{9}{5000} = \frac{73}{15000},$$

giving (8).

For the exact schedule $N = \lceil 2/e^2 \rceil$, one has $d = e^2 + O(e^4)$. Formula (15) gives

$$\delta = ed/2 + (7/4)e^2 d + O(e^3 d + e^2 d^2) = e^3/2 + O(e^4).$$

By the first three estimates in Lemma 5.1, the unpadded maximum-cap secant sum differs from $J(e)$ by $O(e^2 d)$. The coefficient of its linear padding term is

$$\frac{1}{2}\sum_i d\widehat{M}_i g_i^{\mathrm{sec}} = 1 + O(e^2),$$

since $\widehat{M}_i = 1 + O(e^2)$, $g_i^{\mathrm{sec}} = 1 + O(e^2)$ uniformly, and $\sum_i d = 2$. The quadratic padding term is $O(\delta^2)$. Together with Lemma 4.1, these facts prove (9). □

In physical angular notation $\tau = 2e$, the schedule bound becomes

$$R_{\mathrm{sec}} \le \frac{1}{2} - \frac{\tau^2}{240} + \frac{\tau^3}{16} + \frac{9}{32}\tau^4, \qquad N \ge \left\lceil 8/\tau^2 \right\rceil.$$

The coefficient conversions matter when comparing statements written in different source coordinate systems.

6. An exact finite example and its arithmetic certificate

Set $e = 1/10$, $N = 10000$, and $d = 1/5000$. Then

$$M_* = 1.01005625, \qquad \delta = 0.00001400626791979166\ldots < 0.00001401.$$

The definition of δ is the exact rational expression (15), not a decimal truncation. Its maximum axis tangent is $(\sqrt{3}/10)(9999/10000)$.

Even without computing the finite sum, the estimates already proved give

$$R_{\mathrm{sec}}(1/10, 10000) < 0.499929. \tag{35}$$

Indeed, the upper bound for $J(e)$ in (28) is less than 0.499911064, the first term of (29) is less than 0.0000028, its last term is less than 0.000000000026, and $M_*\delta + \delta^2/2 < 0.000014164209$. Their sum is below the claimed value. This analytic estimate is distinct from the tighter exact-sum enclosure (10).

6.1. **Finite rational bounds.** For $m \geq 0$, define the rational Taylor polynomial

$$T_m(z) = \sum_{k=0}^{m} (-1)^k \frac{\binom{2k}{k}}{4^k} z^k.$$

Taylor's integral remainder has the sign of $(-1)^{m+1}$ for $(1+z)^{-1/2}$ on $z \geq 0$. Hence, for every nonnegative z,

$$T_9(z) \leq (1+z)^{-1/2} \leq T_8(z). \tag{36}$$

For rational e, put $D_i = A_i^2 + 2c_i^2$ in the exact finite construction. The two rational numbers

$$L_N = \frac{d}{4} \sum_i M_i^2 T_9(e^2 D_i), \qquad U_N = \frac{d}{4} \sum_i M_i^2 T_8(e^2 D_i) \tag{37}$$

satisfy $L_N \leq R_{\mathrm{sec}}(e, N) \leq U_N$ by (16) and (36). Thus the numerical enclosure reduces to a finite rational calculation with specified polynomials and no floating-point acceptance criterion.

There is no need to materialize all N terms. On positive-midpoint cells the cap maximum is $C_e(c_i + d/2)$, so both the cap and the squared axis norm are polynomials in the index. For even N, the positive midpoints are $(2j-1)/N$, $1 \leq j \leq N/2$; for odd N, they are $2j/N$, $1 \leq j \leq (N-1)/2$, with a separately included central cell. The lower polynomial in (37) has degree at most 44 in j, and the upper polynomial has degree at most 40. Reflection accounts for the negative cells. The exact power sums $S_r(m) = \sum_{j=1}^{m} j^r$ can be evaluated recursively from

$$S_0(m) = m, \qquad S_r(m) = \frac{(m+1)^{r+1} - 1 - \sum_{k=0}^{r-1} \binom{r+1}{k} S_k(m)}{r+1} \quad (r \geq 1).$$

This recurrence is simply the telescoping sum of $(j+1)^{r+1} - j^{r+1}$. It makes (37) a constant-degree rational certificate whose integer lengths grow with $\log N$ for fixed rational e.

The separate implementation `verification/secant_verify.py` evaluates these bounds with exact fractions. It gives the following outward-rounded decimal enclosures:

e	Cylinders $2N$	Certified normalized area interval
1/10	20,000	[0.499921585997001674, 0.499921585997001774]
1/100	40,000	[0.499998863418315727, 0.499998863418315728]
1/1000	4,000,000	[0.499999983836322736, 0.499999983836322737]

In particular, the first row proves (10) and completes Theorem 1.3. The exact rationals in (37), rather than the displayed decimal precision, are the underlying certificate.

6.2. **What the independent verifier checks.** The acceptance proof reconstructs the secant interpolation, the coupled elimination, the polynomial identity (21), and the exact perturbation before estimating it. It checks the rational inequalities in the finite error bounds and evaluates (37). It uses the shared exact polynomial and power-sum primitives in `verification/verify.py`, but does not import the cylinder generator into those acceptance calculations. For $N = 1, 2, 3, 7$, the resulting sums are also checked against a direct sum of rational integer-square-root enclosures.

The generator is imported separately for 36 representative output checks covering both families, $e = 1/100, 1/10$, and $N = 1, 2, 3, 10000$. These check the axes, intercept vertices, padding, parallel displacement under projection, perpendicularity, positive Gram determinants, and exact base areas. Coordinates are stored as $(x, y, z/\sqrt{2})$, with metric $\mathrm{diag}(1, 1, 2)$; thus this check retains the physical metric rather than treating the scaled coordinates as Euclidean. These finite output checks supplement the universal argument in Section 3; they do not replace it with sampled coverage.

The indexed constructor returns any one of the $2N$ cylinders from its index, using exact rational scaled coordinates and the exact area expression. For fixed rational e it needs a constant number of rational values with bit lengths $O(\log N)$, rather than an explicit list of every cylinder.

6.3. **A larger proved range for a source-form cap.** At $e = 1/10$, the source-form compatible cap with $\tau = 1/5$ and $\eta = 1/1000$ has twice its physical radial cutoff equal to

$$R_L(q) = 1 + e^2\{h(q) + 1/125\}.$$

Its excess over the quartic cap is independent of q:

$$R_L(q) - C_e(q) = \frac{19}{800000} = 0.00002375 > \delta(e, 1/5000).$$

Remark 3.4 therefore proves a closed-body cover with this continuous cap and the same 10,000 secant intervals. Its cell-maximum triangular enlargement also covers. This is a new admissibility proof at these parameters, rather than an assertion that they met the older paper's sufficient thresholds. The present triangular cap is smaller on every cell than that enlarged source-form cap and, with identical axes, has strictly smaller cost.

7. Comparison with the identified published ranges

The finite benchmark allows at most 20,000 cylinders. The maximum tangent is bounded by $(\sqrt{3}/10)(9999/10000)$ from the corresponding horizontal equality edges. The radial and generic estimates below allow even the slightly larger tangent budget $\sqrt{3}/10$, so also apply at this benchmark. The comparison concerns the identified source recipes on the explicit sufficient ranges and count conventions in their proofs. A sufficient threshold is not treated as a necessary condition for all geometrically valid covers.

Published recipe and convention	Obstruction at the benchmark
Radial linear, cubic, fractional, and square-root caps on their explicit source ranges; arbitrary partitions and tags, actual projected bases or larger enclosures	$R \geq 1/2 - 1/23328000 > 0.49999995$, allowing retuned source tilt.
Quantitative angular maximum caps, including the retained-mesh sufficient budget and the cubic-buffer family	$R \geq 1/2 - 1/93312000$.
General compatible angular family on the explicit proof witness	$R \geq 1/2 - 1/5529600000$; the full list also needs at least 36,864,000 cylinders.
Separate direct endpoint and squared constructions	At least 392,000,002 and 800,000,000 cylinders, respectively.
Alternative local Cartesian certificate	Any resulting complete realization has $R > 0.499999999996$.
Rational or singular full retained-grid recipe	$R \geq 637/1250 = 0.5096$.
Rational product/logarithmic recipe with its published saving-error allocation	More than 20 million selected main cylinders in one family, including after finite-subcover selection.
Nilpotent recipe with its prescribed apex lists	At least 34,134 cylinders from the apex lists alone.
Shortened beams with the prescribed strip	A strip axis deviates by more than $\arctan(1/3)$, outside the angle budget even after tetrahedron symmetries.

Every listed area floor exceeds the new certified upper bound 0.499922. The table summarizes proved comparisons in the public companion notes: the explicit proof-range audit, the radial and angular actual-area comparison, and the generic realization and count comparison. Those notes include the complete derivations and the precise source locations; their geometric count arguments are not replaced here by a claim about asymptotic coefficients.

For clarity, two features of the comparison merit spelling out. First, the general compatible angular theorem uses the explicit witness [5]

$$E \ge 0, \qquad 0 < \eta \le 1/480, \qquad 0 < \tau \le 1/2, \qquad (10+4E)\tau \le \eta. \tag{38}$$

Its source area estimate is

$$\left| R - \left(\frac{1}{2} + (2\eta - 1/240)\tau^2 \right) \right| \le (6E+2)\tau^4.$$

The source's extra expansion condition $E\tau^2 \le 1/16$ follows from (38). Since $6E+2 \le 2(10+4E)$ and $\tau \le 1/2$, the positive $2\eta\tau^2$ absorbs this remainder. Also $\tau \le 1/4800$. This proves the table's corresponding area floor. On each cell the constant angular coefficient must approximate $q/2$ with error at most $E\tau^2$ at both endpoints. The width is therefore at most $4E\tau^2$. Positive-width cells force $E > 0$, and their full two-family count is at least

$$\frac{1}{E\tau^2} \ge \frac{(10+4E)^2}{E\eta^2} \ge \frac{160}{\eta^2} \ge 36{,}864{,}000,$$

where $(10+4E)^2 - 160E = (4E-10)^2 \ge 0$. The separate direct endpoint and squared certificates use their own ranges, rather than being silently assigned (38).

Second, immediate improvements of the older angular proof belong in the baseline. Retaining its actual mesh Δ gives the sufficient condition

$$2(1-\tau)\eta \ge \frac{7}{4}\tau + \frac{\Delta}{4\tau} + \frac{\Delta}{4}.$$

On the same count schedule, $\Delta \sim \tau^2/4$ gives a canonical e-cubic coefficient $29/2$, and $\Delta = o(\tau^2)$ gives coefficient 14. The coarser substitution $\Delta \le \tau^2$ gives coefficient 16. These are stronger prior-art corollaries than choosing only the larger printed cubic coefficient. The table's actual-area floor includes this optimized retained-mesh condition, not just one favorable printed example.

None of these comparisons asserts that every possible witness of an existential smallness theorem obeys the exhibited numerical witness. Nor do they cover redesigned tile deletion, different product shapes, new error allocations, or new admissibility proofs for generic methods. The cap extension in Section 6 itself demonstrates why that distinction is necessary. There is no global lower bound for all tetrahedron cylinder covers in this note.

8. Supplementary results, reproduction, and verification status

The repository also contains a different finite realization using frozen radial sweeps. Its complete proof is in `THEOREM.md`, with details in the finite-triangle optimality note. Within its stated class—symmetric intercept triangles containing each full constant-maximum-cap frozen sweep—the enclosure is optimal and its count schedule has sharp cubic coefficient 2. These are supplementary results whose proofs are in the linked files, not additional theorems proved in this manuscript. The secant triangles used here do not contain each cell's full continuously tilted sweep, so that restricted lower bound does not apply. Sharpness of the constant $9/16$ within the constant-quartic continuous cap family is likewise proved separately in the cap analysis; global cap optimality is not asserted.

From the repository root, one command verifies all source pins and both exact verification outputs:

```
python3 topics/cylinder-covering/reproduce.py
```

An optional query returns one indexed cylinder:

```
python3 topics/cylinder-covering/secant_construction.py
    -epsilon 1/10 -n 10000 -index 0
```

The two displayed lines form one command. Only the Python standard library is required. The independent certificate report and saved secant verification output document the checked identities, inequalities, and outward intervals.

This package was developed with AI assistance using OpenAI Codex. The evidence for the new theorem consists of the universal written proof and the identified exact-arithmetic checks. It is not a claim of human peer review. The new theorem has not been formalized or kernel checked in Lean. Pinned Lean files are evidence for the cited prior-art definitions, proof witnesses, and area formulas; their presence is not a formal verification of the new quartic/secant theorem.

References

[1] OpenAI, *Finite angular cylinder covers below the half-area bound*, 27 September 2026, OpenAI math family 100. Pinned manuscript and source files.

[2] OpenAI, *Finite triangular approximation of radial sweeps*, 27 September 2026, OpenAI math family 100. Pinned manuscript and source files.

[3] OpenAI, *Finite cylinder approximation of ruled sets*, 27 September 2026, OpenAI math family 100. Pinned manuscript and source files.

[4] OpenAI, *Slope-field perturbations of the two-cylinder covering*, 27 September 2026, OpenAI math family 100. Pinned manuscript and source files.

[5] OpenAI, *Cylinder-covering formal source files*, repository revision `1581d4cbdd69a0faf34b2089614fe2a814316a6a`. The pinned local general coverage, assembly and expansion, admissibility definitions, and physical area identity give the explicit compatible-cap witness discussed in Section 7; other source locations are identified in the proof-range audit.